RiskLens: Transaction Risk Scoring with Behavioral Patterns and Adaptive Anomaly Identification

A PROJECT BASED LEARNING (PBL) REPORT

Submitted by

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Submitted in partial fulfillment of the requirements for

the

Project-Based Learning component of Java Programming

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



CHENNAI INSTITUTE OF TECHNOLOGY



(Autonomous)

Affiliated to Anna University, Chennai

October - 2026

Vision of the Institute:

To be an eminent centre for Academia, Industry and Research by imparting knowledge, relevant practices and inculcating human values to address global challenges through novelty and sustainability.

Mission of the Institute:

IM1. To creates next generation leaders by effective teaching learning methodologies and in still Scientifics park in them to meet the global challenges.

IM2. To transform lives through deployment of emerging technology, novelty and sustainability.

IM3. To inculcate human values and ethical principles to cater the societal needs.

IM4. To contributes towards the research ecosystem by providing a suitable,

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Vision of the Department:

To Excel in the emerging areas of Computer Science and Engineering by imparting knowledge, relevant practices and inculcating human values to transform the students as potential resources to contribute innovatively through advanced computing in real time situations.

Mission of the Department:

DM1. To provide strong fundamentals and technical skills for Computer Science applications through effective teaching learning methodologies.

DM2. To transform lives of the students by nurturing ethical values, creativity and novelty to become Entrepreneurs and establish start-ups.

DM3. To habituate the students to focus on sustainable solutions to improve the quality of life and the welfare of the society.

PROGRAM EDUCATIONAL OBJECTIVES (PEO):

PEO 1: Contribute to the industry as an Engineer through sound knowledge acquired in core engineering to develop new processes and implement the solutions for industrial problems.

PEO 2: Establish an organization / industry as an Entrepreneur with professionalism, leadership quality, teamwork, and ethical values to meet the societal needs.

PEO 3: Create a better future by pursuing higher education / research and develop the sustainable products / solutions to meet the demand.

PROGRAM OUTCOMES (POS):

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOS):

PSO 1 : Analyze, design and develop solutions in the areas of Business Process Management to build the quality products for industry and social needs.

PSO 2 : Innovate ideas and solutions for real time problems in the field of Software Engineering and Mobile applications by adapting the emerging technologies and tools.

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BONAFIDE CERTIFICATE

This is to certify that the Project–Based Learning report titled " RiskLens: Transaction Risk Scoring with Behavioral Patterns and Adaptive Anomaly Identification " is a Bonafide record of work carried out by SYAMRIT ESWARAN GURUSAMY , 2104251041018 of the Department of Computer Science and Engineering, Chennai Institute of Technology, as part of the continuous, mentor–guided Project-Based Learning (PBL) component of the Java Programming course during the academic year 2026–2027 under my supervision.

SIGNATURE
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Submitted for the final review held on …….

INTERNAL EXAMINER

DECLARATION

We jointly declare that the PBL report on **" RiskLens: Transaction Risk Scoring with Behavioral Patterns and Adaptive Anomaly Identification "** is the result of original work done by us and best of our knowledge, similar work has not been submitted to **"ANNA UNIVERSITY CHENNAI"** for the requirement of Degree of **BACHELOR OF ENGINEERING**. This PBL report is submitted on the partial fulfillment of the requirement of the award of Degree of **COMPUTER SCIENCE AND ENGINEERING**.

Signature

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SYAMRIT ESWARAN GURUSAMY

S HARSHAVARTHANAN

Place:

Chennai

Date:

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We wish to express our sincere gratitude to our honourable Chairman **SHRI.P. SRIRAM** for providing immense facilities at our institution.

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